

Executive Summary

AI-Driven Data Research in Quantitative Finance

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IEDA4920 Final Year Project

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Final review-submission package

Project Overview

This repository contains the final review-submission package for the IEDA4920 Final Year Project, *AI-Driven Data Research in Quantitative Finance*. The project develops an end-to-end framework that converts SEC Item 1A *Risk Factors* disclosures into hierarchical textual-risk exposures, semantic peer networks, topology-only ST-GAT forecasts, econometric diagnostics, and academic portfolio-backtesting outputs.

Annual 10-K risk paragraphs are extracted from EDGAR filings, mapped into a dynamic macro-meso taxonomy, aggregated into firm-year exposure vectors, and converted into annual semantic peer graphs. These graphs are used to test whether disclosure-implied peer topology improves one-year-ahead return and volatility forecasts beyond each firm’s own financial and market features.

Core Research Contribution

The central modeling contribution is a strict graph-topology ablation. The full ST-GAT model uses

$$A_t = I + A_{risk,t},$$

while the identity baseline uses

$$A_t = I.$$

This design isolates the incremental value of textual-risk peer topology. If the full graph improves over the identity baseline, the gain is attributable to semantic peer links rather than to a different feature set or a weaker model family.

Empirical Design

The forecasting panel covers risk years 2006–2024. The chronological split is:

Stage	Years	Role
Training	2006–2016	Model fitting, feature scaling, and imputation statistics
Validation	2017–2020	Hyperparameter selection and early-stopping signal
OOS review panel	2021–2024	Final forecast and portfolio diagnostics

Each firm-year node uses 12 backward-looking financial and market features, standardized using training-year statistics only. Features and risk topology are aligned to the July-to-June feature window, while forward return and volatility targets are measured over the following July-to-June window.

Main Verified Results

The finalized report emphasizes the Meso-level graph as the strongest verified specification. On the exported 2021–2024 OOS review panel, Meso ST-GAT improves the identity baseline as follows:

Metric	Identity baseline	Meso ST-GAT
Averaged RMSE	0.3345	0.3001
Averaged MAE	0.2314	0.1991
Averaged Spearman rank	-0.0139	0.1701
Volatility RMSE	0.2240	0.1565
Volatility Spearman rank	-0.1351	0.1770

The clearest improvement is in volatility forecasting, where granular textual risk topology substantially reduces forecast error and reverses the rank correlation from negative to positive.

Graph Interpretation

Graph-density diagnostics support the interpretation that semantic resolution matters. During the OOS period, the Macro graph retains roughly 3,900–4,460 off-diagonal directed edges per year, with density around 5.1%–5.5%. The Meso graph retains only 38–118 off-diagonal directed edges per year, with active-node density below 0.14%.

The report therefore frames the Meso result as evidence that granular risk categories create sparse, high-confidence semantic peer links, rather than merely smoothing information through a dense network.

Portfolio Validation

The portfolio extension is presented as academic validation, not as live-trading evidence. Annual Meso ST-GAT forecasts are held fixed over their July-to-June forecast window, while portfolios are rebalanced monthly with 10 bps transaction costs, 5 bps slippage, and SPY as benchmark.

The strongest return-only long-short strategy reports:

Measure	Value
Annualized return	7.93%
Annualized volatility	7.47%
Sharpe ratio	1.06
Sortino ratio	2.01
Maximum drawdown	-6.90%
Beta versus SPY	0.24
Top-minus-bottom spread	17.69% annualized
Nominal spread p-value	0.034

The identity-baseline return-only signal has a smaller and insignificant 6.62% annualized top-minus-bottom spread. This supports the interpretation that semantic peer topology improves cross-sectional ranking quality, while the report avoids presenting the result as investment advice or live-trading deployability.

Claim Safety and Limitations

The final package is deliberately conservative about unsupported evidence. Taxonomy-coverage percentages, AAPL firm-level migration claims, DAV/EGARCH-X BIC comparisons, Fama-MacBeth tables, and SAR/network regression claims are not used as headline findings unless their source tables are regenerated and added to the output manifest.

The available DAV peak diagnostics are retained only as event-local context: 93 figures were parsed, with 20 positive improvements, 73 negative improvements, and mean improvement of -2.40%. These diagnostics do not establish broad DAV dominance.

Repository Status

Before GitHub cleanup, the complete local artifact set passed the lightweight validation suite with 45 passes, 2 warnings, and 0 failures as of 2026-04-27. The warnings are research caveats rather than code crashes: four financial feature values require downstream train-period median imputation, and risk-year 2024 has a forward target window ending on 2026-06-30.

The committed GitHub repository is a lightweight code, report, presentation, and documentation package. Large raw data, intermediate files, generated outputs, logs, and the local virtual environment are excluded from version control and preserved outside the repository as local artifacts.